

ASSESSMENT 3 BRIEF	
Subject Code and Title	MLN 601 Machine Learning
Assessment	Machine Learning Project
Individual/Group	Individual
Length	2,000 words (+/-10%)
Learning Outcomes	The Subject Learning Outcomes demonstrated by successful completion of the task below include: a) Evaluate and compare the key concepts in machine learning. b) Apply learning algorithms to perform machine learning tasks. c) Implement practical machine learning: data pre-processing, analysis, model selection, and interpret the results. d) Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders
Submission	12-Week Duration: Due by 11:55pm AEST/AEDT, Wednesday of week 12 6-Week Duration: Due by 11:55pm AEST/AEDT, Wednesday of week 6
Weighting	40%
Total Marks	100 marks

Task Summary

Bike sharing systems are poised to become a major mode of transportation. The prediction problem for determining how many bikes will be present at a station is important to both travellers and the entrepreneurs running the systems. This Assessment requires you to develop at least two models using different modelling algorithms to predict the bike demand from a real-world data set.

Please refer to the Task Instructions (below) for further details on how to complete this task.

Context

This assessment requires you to make predictions based on at least two different techniques in machine learning (ML) using real-world data and compare the performance of your predictions. As a professional, you will often be expected to perform similar tasks using suitable data sets and you must be open to trying different techniques rather than having a bias towards just one approach.

For this Assessment, the practice data sets are available from the UCI ML repository, which contains nearly 500 real-world data sets (<http://archive.ics.uci.edu/ml/index.php>). The data sets for this investigation are the Bike Sharing data sets (<https://archive.ics.uci.edu/ml/machine-learning-databases/00275/>).



The data sets contain daily counts of bike rentals from the bike sharing company Capital-Bikeshare15 in Washington D.C., along with weather and seasonal information. The goal of your model is to predict how many rental bikes will be available on the street for a given day and weather forecast.

The selection of models has been left to you, but could include a linear model, support vector machine, nearest neighbour, random forest, gradient boosting or any other learning models available in scikit-learn. For the error measurement, use the mean absolute error.

You must consider the Cross-Industry Standard Process for Data Mining (CRISP-DM; Chapman et al 2000) from the outset. You will commence with the business understanding step to ensure the correct framing of the problem (e.g., to predict the demand for bicycles on a given day and weather forecast). In this stage, you should also identify any constraints and assumptions. The point of commencing the analysis and modelling with a business understanding is to provide you with an opportunity to consider some of the variables influencing demand rather than jumping straight into the exploratory analysis.

Task Instructions

Multiple activities are required to complete this assessment task. Follow the steps of the CRISP-DM model using the template (CRISP-DM Template.ipynb) to document and develop your ML model.

Stage 1: Business Understanding

1. This section serves as an introduction. You should write a clear and concise narrative expressing what you are trying to achieve. Think in terms of ML (e.g., the prediction algorithm, the data set selected, what you are seeking from the data set and how you intend to understand the value of your prediction capability).
2. Assess the current situation.

Stage 2: Data Understanding

1. Acquire the relevant bike data set from the UCI repository for your prediction model (<https://archive.ics.uci.edu/ml/machine-learning-databases/00275/>). Explicitly specify the data source by providing a specific link and the name of the data set (e.g. Bike Sharing Dataset) and the method of acquisition (e.g., direct from the URL or a download of the .csv file). The steps taken need to be clearly stated.
2. Read this data set into your Notebook.
3. Describe the data set inclusive of variables, units and levels.
4. Verify the data quality by analysing the data set for structure and missing data.
5. Conduct an initial data exploration using data visualisation, reporting and querying the data.
6. Use the pairplot function in seaborn to determine the relationship, if any, between the variables. Include the output or the visualisation of the pairplot function in your Notebook and comment on it.



Stage 3: Data Preparation

1. Select the data that you will use for the analysis.
2. Clean the data you have selected to improve the quality of the data.

Stage 4: Modelling

1. For this Assessment, you are required to consider at least two models.
2. Import the models into your code.
3. Record any modelling assumptions.
4. Create a training and testing data set (e.g., 25% testing and 75% training).
5. Build different prediction models to run over the data set and predict the demand for a given day and weather forecast.
6. Record the model parameter settings, including your rationale for the choice of values and the actual model generated.
7. Assess the models according to the performance measurement.
8. Revise any parameter settings for subsequent model runs. Document all revisions until the best model is reached.
9. Comment on the model's most predictive features.

Stage 5: Evaluation

1. Assess the ML results. Ensure you include a statement as to whether the selected model meets the evaluation criteria.

Stage 6: Deployment

1. For this Assessment, you are not required to deploy your model. For this stage, simply include any lessons that you learned and that you wish to share regarding the things that went right and wrong, the areas in which you did well and in which you could improve. You can also detail any of your other experiences in completing this Assessment.

Submission Instructions

You are expected to submit two files: 1) The Notebook file with an extension .ipynb; and 2) The direct PDF download via LaTeX from the Notebook. Name the files as follows:

MLN601LastNameFirstNameBrief3.ipynb; and

MLN601LastNameFirstNameBrief.pdf.

IMPORTANT NOTE: Submit a text file (.txt) that includes the code(s) of your programs in your submissions.

Submit the files via the Assessment link in the main navigation menu in **MLN601—Machine Learning**.

The Learning Facilitator will provide feedback via the Grade Centre in the LMS portal. Feedback can be viewed in My Grades.

References

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C. & Wirth, R. (2000). *CRISP-DM 1.0—Step-by-step data mining guide*. Copenhagen, Denmark: SPSS Inc. <https://www.the-modeling-agency.com/crisp-dm.pdf>

UCI (2019). *Machine learning repository*. Irvine, CA: University of California, School of Information and Computer Science. <http://archive.ics.uci.edu/ml>

Academic Integrity Declaration

I declare that except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>

I am aware that I need to keep a copy of all submitted material and their drafts, and I will do so accordingly.

Assessment Rubric

Assessment Attributes	Fail (Yet to Achieve Minimum Standard) 0– 49%	Pass (Functional) 50–64%	Credit (Proficient) 65– 74%	Distinction (Advanced) 75– 84%	High Distinction (Exceptional) 85– 100%
Business Understanding 10%	Not really attempted, incomplete and difficult to comprehend.	Demonstrates an attempt to frame the problem; for example, to predict rental bike usage to help with overall planning and meet demand.	Contains a clear explanation of the motivation to undertake ML (e.g., prediction). Further, the problem statement recognises the ability of bike sharing systems to acquire rental bike usage data.	The entire introduction to the report is clear, provides a narrative on the bike sharing system and points out the ability to determine the variables significantly affecting bike usage.	Complete, readable and easy to understand and includes a commentary on bike sharing, prediction ability and identifies the bike sharing variables. Demonstrates a desire to understand more about bike sharing. Includes citations to any additional readings.
Data Understanding 20%	Entirely incorrect and not completely attempted. Totally lacking any interpretation or discussion of the independent and dependent variables. Fails to specify the source of the data set or the number of observations.	Partially correct but missing plots and poorly labelled. Provides limited comments on the source of the data and variables. Does not explicitly state the dependent variable. Specifies the original source of the data set.	The plots and analysis are partially complete, including the correlation analysis; however, there are minor omissions in interpretation. Makes clear the source of the data and the dependent variable selected. For example, “cnt”, the total count of rental bikes.	The analysis describing the data and plots is complete and includes a visualisation and plots the variables to cnt. Not all of the plots are suitable. Some useful plots worth highlighting are: season/month/week/hour against the total rental bikes, weather against the total bikes, temperature, humidity and windspeed against the total bikes and casual versus registered bikes.	The data understanding has been explicitly tied to the data analysis and context using all the plots: season/month/week/hour against the total rental bikes, weather against the total bikes, temperature, humidity and windspeed against the total bikes and casual versus registered bikes.

			Some useful plots included to understand the data, such as season/month/week/hour against the total rental bikes, weather against the total bikes, temperature, humidity and windspeed against the total bikes and casual versus registered bikes. Some missing plots, but overall sound.	humidity and windspeed against the total bikes and casual versus registered bikes.	Conveys information and includes adequate references to the data set.
Data Preparation 10%	Comments missing and appears to be incomplete. No attempt to create a training test data set. No comments made about the correlation matrix, which is likely absent.	Partially complete; uses training/testing data set (e.g., 2011/2012). The correlation matrix visualises the variables; however, there is little or no commentary on the positive or negative correlations.	Covers data quality and discusses the training/testing data set split. The positive and negative correlations are discussed. The highest correlation variable with cnt is noted.	The discussion of the data preparation steps are made explicit as is the training/testing split. The positive and negative correlations are discussed. The highest correlation variable with cnt is noted. Any negative correlation is stated and discussed.	The discussion is clear in relation the correlations. And appears to be correct in relation to the training/testing data. Provides comments in relation to selection of daily data over hourly data to avoid hourly bias. May indeed provide further comments with regard to estimating missing data.
Modelling and Evaluation 30%	Overly simplistic and incomplete. Failed to provide at least two different models.	An appropriate analysis of two models with the ML code. Lacks model assessment comments.	The modelling generation and assessment is partially complete. Minor errors or omissions.	The modelling is mainly correct for at least two models in relation to performance review but includes some minor errors or lacks some depth.	The modelling is complete in relation to the assessment and includes informative comments that aid interpretation.

		Did not provide metrics MSE or R ² for any of the models.	Some attempt to review the performance of two models. Provides metrics (MSE & R ²).	Highlights training and validation with metrics for at least two models.	Went beyond just two models to include an ensemble. Proposes next steps to improve the models.
Deployment 10%	No real attempt to draw a conclusion detailing the lessons learned or experiences (no or very little reflection).	The conclusion is relatively well structured but fails to detail the lessons learned or discuss the experiences.	The conclusion details the lessons learned and is mostly clear to follow. Not much else provided in relation to experiences.	The conclusion details the lessons learned and the experiences.	The conclusion clearly details the lessons learned and experiences. The conclusion has some depth (e.g., suggests what could be done better next time). Some additional references on building a bike sharing system network have been considered.
Deliverables: Jupyter code Pdf 20%	Insufficient comments made in the markdown reporting. Jupyter Notebook file or pdf not provided. Did not attempt to use the template provided.	Notebook file requires major changes to run. Markdown is hard to follow and confusing. The presentation is similarly hard to follow, as it is dependent on the markdown.	Notebook runs with minor changes. Markdown is reasonable and can be followed. Pdf output is similar to the Notebook. Output from at least two models.	All files run and the Notebook reproducing the submitted files.	The markdown report documents the entire end-to-end process and makes full use of the CRISP-DM stages as highlighted in the template. The Notebook file runs with no errors and matches the pdf.

					The output touches all key the stages of the CRISP-DM and concludes with comments on the student's own experiences and suggestions for improvement.
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The following Subject Learning Outcomes are addressed in this assessment	
SLO a)	Evaluate and compare the key concepts in machine learning that are crucial in initiating collaborations in industry and academia.
SLO b)	Apply learning algorithms to perform machine learning tasks.
SLO c)	Implement practical machine learning: data pre-processing, analysis, model selection and interpret the results.
SLO d)	Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders